

Learning and Decision Trees

- **Based on a set of attributes as input, predicted output value, the *decision* is learned**
 - it is called *classification* learning for discrete values
 - *regression* for continuous values
- **Boolean or binary classification**
 - output values are true or false
 - conceptually the simplest case, but still quite powerful
- **Making decisions**
 - a sequence of test is performed, testing the value of one of the attributes in each step
 - when a leaf node is reached, its value is returned
 - good correspondence to human decision-making

Boolean Decision Trees

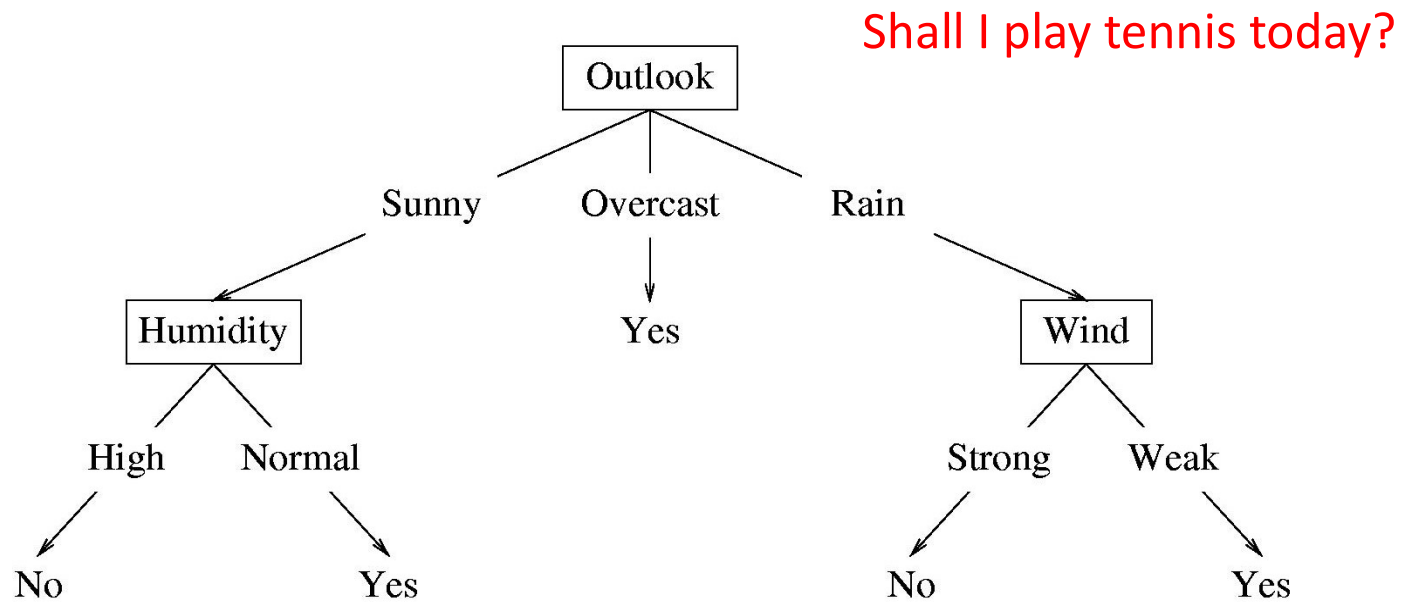
- A decision-tree learning algorithm approximates a target concept using a tree representation
 - Each **internal node** corresponds to a test of the values of attributes
 - Each **Branches** are the possible values of the test
 - every **terminal node** corresponds to a class which is to be returned.
- The attributes are selected in such a way so that the depth of the final tree can be minimized.
- The idea is to pick the attribute that goes towards an exact classification of the examples.
- A perfect attribute divides the example that are all positive or all negative.

Method: Decision trees

- “Divide and conquer”:
Split data into smaller subsets – choose random subset of training example – the subset is called **Window**
- Algorithm builds a decision tree that correctly classify all the example in the window.
- The tree is then tested on the examples outside the window.
- If, all the examples are classified correctly then the algorithm halts.
- Else, add number of training examples to the window and repeat the process.

Decision Tree Hypothesis Space

- **Internal nodes** test the value of particular features x_j and branch according to the results of the test.
- **Leaf nodes** specify the class $h(\mathbf{x})$.



Suppose the features are **Outlook** (x_1), **Temperature** (x_2), **Humidity** (x_3), and **Wind** (x_4). Then the feature vector $\mathbf{x} = (\text{Sunny}, \text{Hot}, \text{High}, \text{Strong})$ will be classified as **No**. The **Temperature** feature is irrelevant.

Explanation-Based Learning

- Learning complex concepts using Induction procedures typically requires a ***substantial number of training instances***.
- We don't need to see dozens of positive and negative examples of fork (chess) positions in order to learn to avoid this trap in the future and perhaps use it to our advantage.
- What makes such ***single-example learning possible***? The answer is knowledge.
- Much of the recent work in machine learning has moved away from the empirical, data intensive approach described in the last section toward this more ***analytical knowledge*** intensive approach.
- An EBL system attempts to learn from a ***single example x by explaining*** why x is an example of the target concept.
- ***The explanation is then generalized***, and then system's performance is improved through the availability of this knowledge.

Explanation-Based Learning

- **EBL programs** as accepting the following as input:
 - **A training example**
 - **A goal concept:** A high level description of what the program is supposed to learn
 - **An operational criterion** - A description of which concepts are usable.
 - **A domain theory:** A set of rules that describe relationships between objects and actions in a domain.
- From this EBL computes a ***generalization of the training example that is sufficient to describe the goal concept***, and also satisfies the operationality criterion.
- Explanation-based generalization (EBG) is an algorithm for EBL and has two steps: **(1) explain, (2) generalize**
- During the explanation step, the domain theory is used to ***prune away all the unimportant aspects*** of the training example with respect to the goal concept. What is left is an ***explanation of why the training example is an instance of the goal concept***. This explanation is expressed in terms that satisfy the operationality criterion.
- The next step is to generalize the explanation as far as possible while still describing the goal concept.